

WASP-10b

R-0869

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_230903 · created 2026-07-28T23:07:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460190.7413 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.135 +/- 0.029
Transit depth (Rp/Rs)^2	1.83 +/- 0.78 [%]
Orbital Inclination (inc)	89.57 +/- 1.21
Airmass coefficient 1 (a1)	1.001 +/- 0.026
Airmass coefficient 2 (a2)	-0.001 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.62 %
Best Comparison Star	#3 - [216, 324]
Optimal Aperture	3.8
Optimal Annulus	7.27
Transit Duration (day)	0.0908 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.135 ± 0.029 vs 0.1592 (deviation 0.83σ)
Duration fit vs published	0.0908 d vs 0.0946 d (deviation 0.9σ)
Mid-time O–C	-2.4 min
Depth SNR	4.7
Residual scatter	2.62 %

Light curve

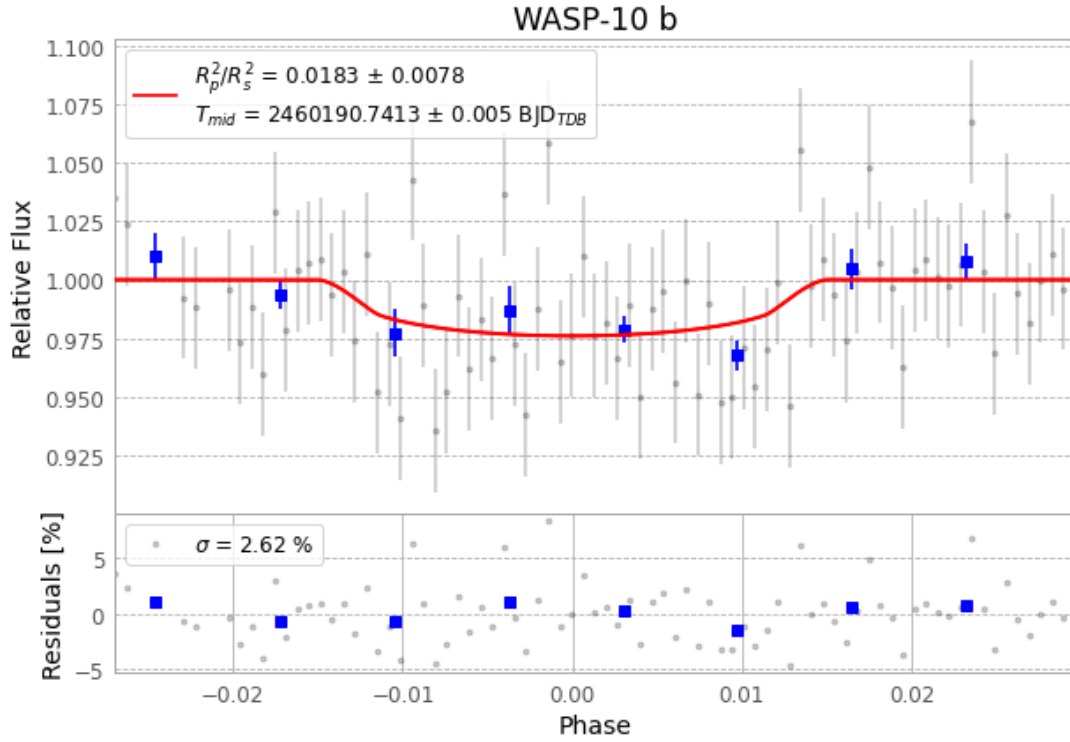


Figure 1 — FinalLightCurve_WASP-10 b_2023-09-02.png (SHA-256 58799382680b9a2c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [250, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2208aa354772ef23...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

85 FITS frames (56 MB), WASP-10230903033915.FITS → WASP-10230903075115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-230903.log (ee023e2b4345...), FinalLightCurve_WASP-10 b_2023-09-02.png (58799382680b...), FinalLightCurve_WASP-10 b_2023-09-02.csv (810ed1094b1d...)

Compute resources

Wall time 385.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 23:07Z.